

Ideation Document

My ideation:

- 1) My code starts with reading all of the four csv file.
 - 2) Then it computes four parameters P, R, E & F .
- P : Student Progress Score, with

$$P = \frac{\text{No. of milestones completed by the students}}{\text{Total no. of assigned milestones}}$$

→ R : Responsiveness Score: basically the avg time taken by the mentor to answer the queries of the mentees

Higher t_{avg} → poor R , lower t_{avg} → better R

↪ $R \propto \frac{1}{t_{avg}}$,

I took $R = \frac{1}{1+t_{avg}}$

why?

for very small t_{avg} R does not be come very large

$R \in [0, 1]$

→ E : Engagement Score, $E = P_1 * \text{avg meetings} + P_2 * \text{avg Code Reviews} + P_3 * \text{avg Messages}$

I took: $P_1 = 0.25$, $P_2 = 0.4$, $P_3 = 0.35$, score: $1 - e^{-E} \in [0, 1]$

$P_2 > P_3 > P_1$: according to me reviewing the correctness of the code saves a lot of time from extensive debugging is the most important factor & in the meetings since there are a many students it is not humanly possible for the mentor to review extensively each mentee.

* In order to normalise the scores among the mentors, in my code I have take the avg values

→ F: Meeting Feedback score: for calculation of the net feedback of all the mentors, instead of taking its mean (which could be unfair) I have taken its standard deviation since it describes the spread of all the ratings

Score calculation: $w_1 P + w_2 R + w_3 E + w_4 F$

$$w_1 = 0.35, \quad w_2 = 0.2, \quad w_3 = 0.3, \quad w_4 = 0.15$$

I chose $w_1 > w_3 > w_2 > w_4$: since the completion of the project is 1st the responsibility of the mentee, then I chose $w_3 > w_2$ since it's more important for the queries to be solved & debugging to be done than how frequently the mentor replies since they also have their own time commitments

→ I have also smoothed the score over the entire time keeping in mind that the current situation has more weightage compared^x to the previous.

And a time decay fn which heavily penalises those who are inactive.

- Then finally I make the csv file & all the mentors who have the same score are given the same rank.